Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

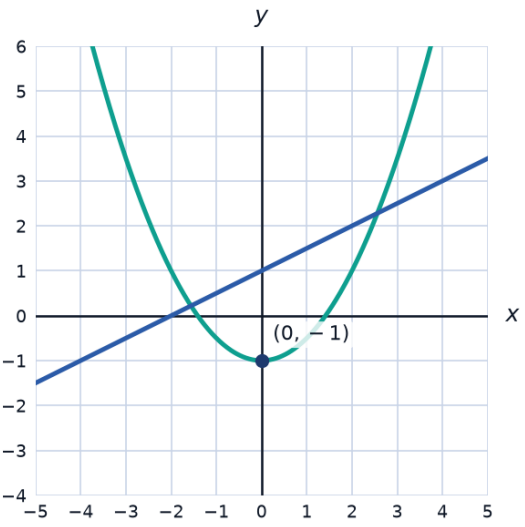
1 [Foundational] • Target: Measures of Central Tendency & Dispersion

A machine's failure probability increases each hour it runs; X is the number of failures in a day. Is X binomially distributed? Show your complete algebraic work.

2 [Foundational] • Target: Probability Rules & Conditional Probability

A discrete random variable X has probability distribution given by

$P(X = 5) = k, P(X = 4) = 5k, P(X = 3) = 3k, P(X = 2) = 4k, P(X = 1) = 2k$, where k is a constant. Find k.



3 [Foundational] • Target: Discrete Probability Distributions

A discrete random variable X has probability distribution given by

$P(X = 6) = 4k, P(X = 2) = 3k, P(X = 3) = 5k, P(X = 4) = 2k, P(X = 1) = 3k$, where k is a constant. Find k.

4 [Foundational] • Target: Binomial Distribution & Exam Applications

A discrete random variable X has probability distribution given by

$P(X = 2) = 4k, P(X = 6) = 3k, P(X = 1) = k, P(X = 4) = 4k, P(X = 5) = 4k$, where k is a constant. Find k.

5 [Foundational] • Target: Measures of Central Tendency & Dispersion

A discrete random variable X has probability distribution given by

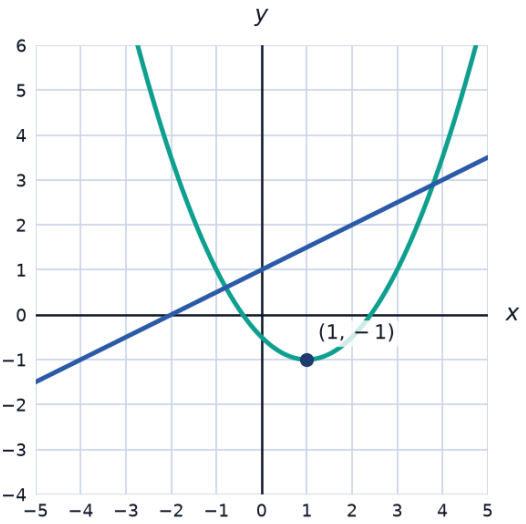
$P(X = 0) = 5k, P(X = 1) = k, P(X = 6) = 4k, P(X = 3) = 2k, P(X = 5) = 5k$, where k is a constant. Find k.

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6 [Foundational] • Target: Probability Rules & Conditional Probability

A discrete random variable X has probability distribution given by

$P(X = 6) = 2k, P(X = 2) = 3k, P(X = 0) = k, P(X = 1) = 3k, P(X = 3) = 4k$, where k is a constant. Find k.



7 [Foundational] • Target: Discrete Probability Distributions

A discrete random variable X has probability distribution given by

$P(X = 0) = 2k, P(X = 3) = k, P(X = 1) = 5k, P(X = 6) = k, P(X = 5) = 5k$, where k is a constant. Find k.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Binomial Distribution & Exam Applications

A discrete random variable X has probability distribution given by

$P(X = 0) = 4k, P(X = 3) = 3k, P(X = 2) = 5k, P(X = 6) = 4k, P(X = 1) = k$, where k is a constant. Find k.

9 [Intermediate] • Target: Measures of Central Tendency & Dispersion

A discrete random variable X has probability distribution given by

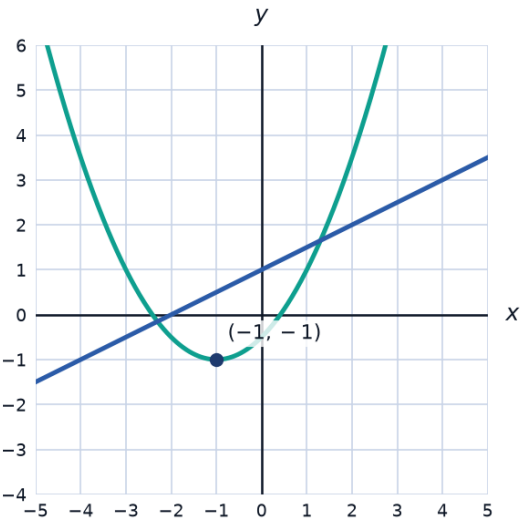
$P(X = 2) = 2k, P(X = 3) = k, P(X = 5) = 2k, P(X = 4) = 4k, P(X = 6) = 4k$, where k is a constant. Find k.

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10 [Intermediate] • Target: Probability Rules & Conditional Probability

A discrete random variable X has probability distribution given by

$P(X=1)=4k, P(X=2)=k, P(X=3)=4k, P(X=4)=k$, where k is a constant. Find k.



11 [Intermediate] • Target: Discrete Probability Distributions

A discrete random variable X has probability distribution given by

$P(X=3)=k, P(X=2)=3k, P(X=1)=k, P(X=4)=3k, P(X=6)=k$, where k is a constant. Find k.

12 [Intermediate] • Target: Binomial Distribution & Exam Applications

A discrete random variable X has probability distribution given by

$P(X=1)=k, P(X=2)=k, P(X=3)=k, P(X=4)=4k$, where k is a constant. Find k.

13 [Intermediate] • Target: Measures of Central Tendency & Dispersion

A discrete random variable X has probability distribution given by

$P(X=1)=k, P(X=2)=k, P(X=3)=2k$, where k is a constant. Find k.

14 [Intermediate] • Target: Probability Rules & Conditional Probability

A discrete random variable X has probability distribution given by

$P(X=1)=3k, P(X=2)=3k, P(X=3)=k$, where k is a constant. Find k.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 3.01 Graphs & Transformations

A discrete random variable X has probability distribution given by

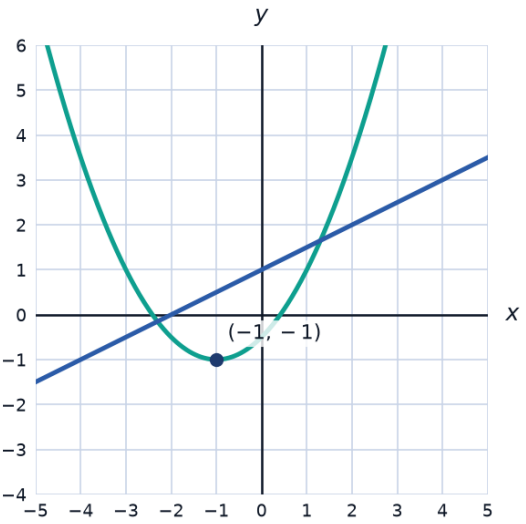
$P(X=1)=k, P(X=2)=3k, P(X=3)=k$, where k is a constant. Find k.

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16 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 4.01 Coordinate Geometry

A discrete random variable X has probability distribution given by

$P(X = 1) = 2k, P(X = 2) = 3k, P(X = 3) = 2k$, where k is a constant. Find k.



17 [Challenging] • Target: Measures of Central Tendency & Dispersion • Cross-Lesson: 5.01 Trigonometry

A discrete random variable X has probability distribution given by

$P(X = 1) = 2k, P(X = 2) = k, P(X = 3) = 2k$, where k is a constant. Find k.

18 [Challenging] • Target: Probability Rules & Conditional Probability • Cross-Lesson: 6.01 Differentiation & Integration

A discrete random variable X has probability distribution given by

$P(X = 1) = k, P(X = 2) = 2k, P(X = 3) = 3k$, where k is a constant. Find k.

19 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 8.01 Functions: Composite & Inverse

A discrete random variable X has probability distribution given by

$P(X = 1) = 3k, P(X = 2) = 2k, P(X = 3) = 2k$, where k is a constant. Find k.

20 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 9.01 Circular Measure

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield ≥ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Measures of Central Tendency & Dispersion

The numbers $4, 7, 9, 12, 18$ have a mean of $\bar{x} = 10$. Find the standard deviation σ correct to 2 decimal places.

a. 4.73

b. 5.21

c. 4.12

d. 3.85

22 [Foundational] • Target: Measures of Central Tendency & Dispersion

A data set of 20 observations has $\sum x = 240$ and $\sum x^2 = 3280$. Find the variance of the data set.

a. 20

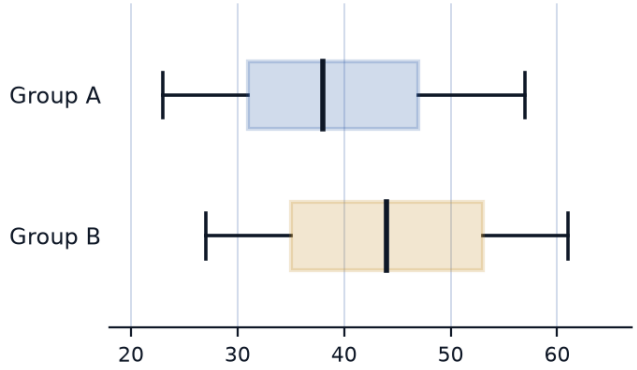
b. 16

c. 24

d. 12

23 [Foundational] • Target: Measures of Central Tendency & Dispersion

In a box-and-whisker plot, the lower quartile is $Q_1 = 24$ and the upper quartile is $Q_3 = 42$. According to the $1.5 \times$ IQR rule, what is the boundary for outliers at the upper end?



a. 69

b. 60

c. 65

d. 72

24 [Foundational] • Target: Probability Rules & Conditional Probability

Two independent events A and B have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cup B)$.

a. 0.70

b. 0.90

c. 0.20

d. 0.65

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25 [Foundational] • Target: Probability Rules & Conditional Probability

Given $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find the conditional probability $P(A \mid B)$.

a. 0.60

b. 0.50

c. 0.30

d. 0.75

26 [Foundational] • Target: Probability Rules & Conditional Probability

A bag contains 4 red balls and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both balls are red?

a. $\frac{2}{15}$

b. $\frac{4}{25}$

c. $\frac{1}{6}$

d. $\frac{1}{5}$

27 [Foundational] • Target: Discrete Probability Distributions

A discrete random variable X has probability function $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. Find the value of k.

a. $\frac{1}{10}$

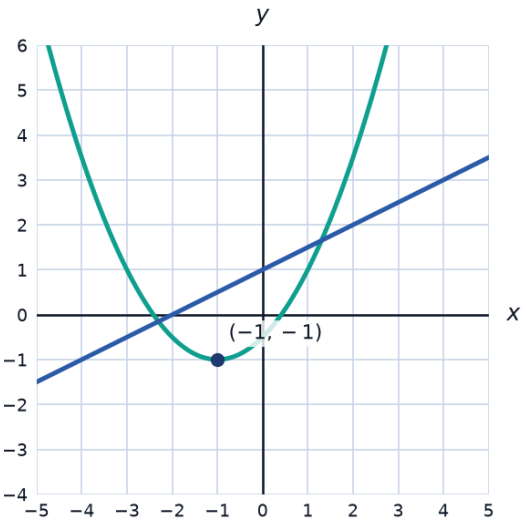
b. $\frac{1}{6}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$

28 [Foundational] • Target: Discrete Probability Distributions

Find the expectation $E(X)$ for the distribution where $P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$.



a. 2.1

b. 2.0

c. 2.3

d. 1.9

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Discrete Probability Distributions**

For a discrete random variable X with $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\mathrm{Var}(X)$?

a. 9

b. 5

c. 16

d. 21

30 **[Intermediate]** • **Target: Binomial Distribution & Exam Applications**

A random variable $X \sim B(8, 0.25)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.3115

b. 0.2076

c. 0.2670

d. 0.1038

31 **[Intermediate]** • **Target: Binomial Distribution & Exam Applications**

If $X \sim B(100, 0.4)$, find the mean and variance of X.

a. Mean $= 40$, Variance $= 24$

b. Mean $= 40$, Variance $= 16$

c. Mean $= 50$, Variance $= 25$

d. Mean $= 40$, Variance $= 4.9$

32 **[Intermediate]** • **Target: Binomial Distribution & Exam Applications**

A multiple-choice quiz has 5 questions, each with 4 options and exactly one correct answer. If a student guesses at random on all 5 questions, what is the probability of passing with at least 3 correct answers?

a. $\frac{53}{512} \approx 0.1035$

b. 0.2500

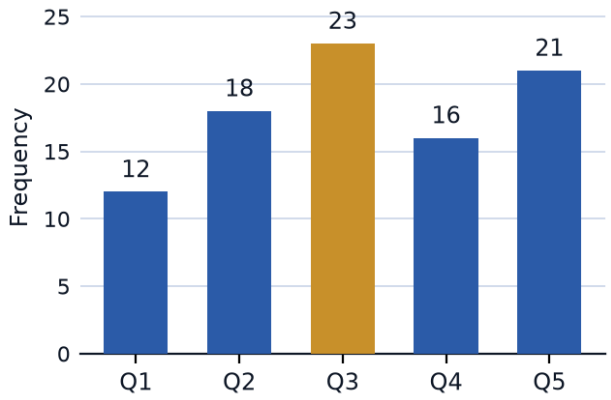
c. 0.0879

d. 0.1563

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33 [Intermediate] • Target: Measures of Central Tendency & Dispersion

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group. | | Car | Bus | Walk | Total | |--------------|-----|-----|------|-------| | Year 1 | 15 | 20 | 10 | 45 | | Year 2 | 25 | 18 | 12 | 55 | | Total | 40 | 38 | 22 | 100 | What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?



a. 0.15

b. 0.33

c. 0.38

d. 0.45

34 [Intermediate] • Target: Probability Rules & Conditional Probability

A bag contains 18 counters, of which 9 are red. A counter is drawn at random. Find the probability that it is red.

a. 1/4

b. 1/2

c. 15/17

d. 7/10

35 [Intermediate] • Target: Discrete Probability Distributions

A discrete random variable X has distribution $P(X=1)=2/5$, $P(X=2)=1/5$, $P(X=3)=2/5$. Given $Y=5X-3$, find $E(Y)$.

a. 6

b. 7

c. 8

d. 13

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36 [Intermediate] • Target: Binomial Distribution & Exam Applications

A discrete random variable X has distribution $P(X=1)=3/10, P(X=2)=2/5, P(X=3)=3/10$. Given $Y=4X-4$, find $E(Y)$.

a. 3

b. 5

c. 4

d. 12

37 [Intermediate] • Target: Measures of Central Tendency & Dispersion

Given $X \sim B(58, 9/20)$, find $E(X)$.

a. 133/5

b. 261/10

c. 251/10

d. 271/10

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Probability Rules & Conditional Probability • Cross-Lesson: 6.01 Differentiation & Integration

Given $X \sim B(46, 9/20)$, find $E(X)$.

a. 207/10

b. 106/5

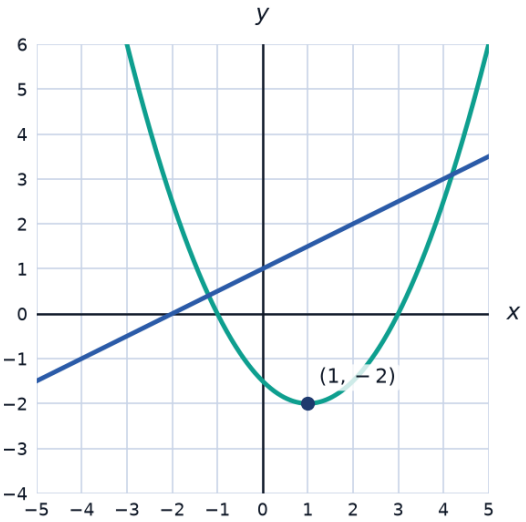
c. 197/10

d. 217/10

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39 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 8.01 Functions: Composite & Inverse

Given $X \sim B(37, 7/20)$, find $E(X)$.



a. 259/20

b. 239/20

c. 269/20

d. 279/20

40 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 9.01 Circular Measure

Given $X \sim B(47, 3/20)$, find $E(X)$.

a. 141/20

b. 121/20

c. 161/20

d. 151/20

41 [Challenging] • Target: Measures of Central Tendency & Dispersion • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Given $X \sim B(75, 11/20)$, find $E(X)$.

a. 167/4

b. 169/4

c. 161/4

d. 165/4

42 [Challenging] • Target: Probability Rules & Conditional Probability • Cross-Lesson: 10.01 Series & Binomial Theorem

Given $X \sim B(75, 13/20)$, find $E(X)$.

a. 191/4

b. 195/4

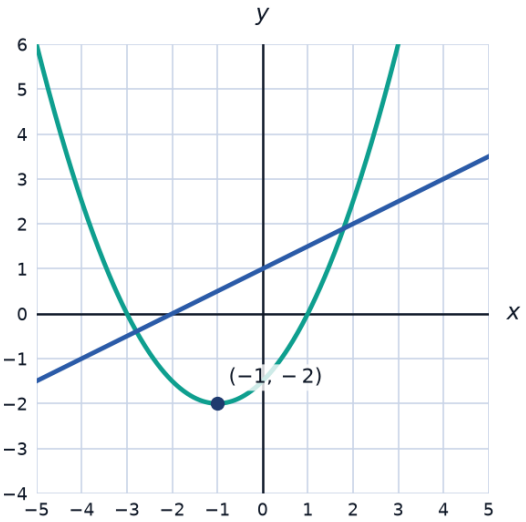
c. 199/4

d. 197/4

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43 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 11.01 Kinematics

Given $X \sim B(17, 1/10)$, find $E(X)$.



a. 27/10

b. 17/10

c. 11/5

d. 7/10

44 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 12.01 Forces & Equilibrium

Given $X \sim B(26, 3/20)$, find $E(X)$.

a. 49/10

b. 22/5

c. 29/10

d. 39/10

45 [Challenging] • Target: Measures of Central Tendency & Dispersion • Cross-Lesson: 13.01 Newton's Laws

Students are surveyed until the first one who owns a car is found; X is the number of students asked. Is X binomially distributed?

a. Yes — but only if $n > 30$

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. No — trials are not independent / probability is not constant

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